

Lilian Laporte

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Education

09.2022–05.2025 **M.Sc. in Robotics, Systems & Controls**
ETHZ – Zürich, Switzerland
Diploma: MSc in Robotics, Systems & Controls (GPA: 5.6/6)

09.2019–07.2022 **B.Sc. in Microtechnics**
EPFL – Lausanne, Switzerland
Diploma: BSc in Microtechnics (GPA: 5.05/6)

08.2021–07.2022 **Exchange Program – Electrical Engineering, Mathematics and Computer Science**
TUD – Delft, Netherlands

09.2012–06.2019 **High school and Secondary school**
St. François de Sales – Ville-la-Grand, France (#1 in France, *Le Monde*, 2015)
Diploma: French Baccalaureate S (mathematical speciality), *with distinction*

Work Experience

01.2026–Today **Research Engineer – RSL, ETHZ – Zürich, Switzerland** (Full-time)
• Develop and optimize robust, efficient ladder-climbing and perceptive locomotion abilities for a quadruped robot, including training in IsaacSim and deployment on real hardware using C++ and ROS.
• Evaluate industrial applications and market potential, aligning robotic capabilities with use cases in plant environments.

02.2024–09.2024 **Robotics Intern – MESH AG – Birr, Switzerland** (Full-time)
• Developed a robotic system to automate rebar stacking, improving efficiency and accuracy for construction suppliers.
• Practical experience in Computer Vision, ROS, and robotic software engineering while enhancing the app's functionality and my full-stack skills.
• **Side project:** RL pipeline in Mujoco to derive an optimal stacking policy based on the shape of the rebar.

08.2022 **Machinist – Rolex, Bracelet Dpt – Plan-les-Ouates, Switzerland** (Full-time)
• Managed a 3-axis CNC for manufacturing bracelet stands and polishing them.
• Firsthand experience in a production chain, understanding key challenges in industrial manufacturing within a multinational company.

10.2021–05.2022 **Research assistant – AIR Lab – Delft, Netherlands** (5h/week)
• Assisted a PhD student in developing code for controlling a robot in the retail supply chain, from warehouses to customers using robotics programming high-level languages.

07.2019 **Electrical Technician – Rolex, Energy Dpt – Plan-les-Ouates, Switzerland** (Full-time)
• Processed career data in the energy sector and finding solutions for electrical issues in a factory of 2000 employees.
• Hands-on experience with a variety of tasks, enhancing my versatility.

Projects

11.2024–06.2025 **Master's Thesis - Robotics Systems Lab (RSL) - ETHZ**
"Guiding Exploration in Fine-Tuning via Simple Trajectory Pre-training for Advanced Skill Learning"
Developed a two-stage learning framework that combines imitation-based pre-training and reinforcement learning for complex robot behaviours, to enable a wheeled quadruped robot to perform stunts (e.g. backflips).

09.2023–02.2024 **Semester Project – IBM/Acoustic Robotics Systems Lab (ARSL) – ETHZ**
"Autonomous Ultrasound-Driven Manipulation of Microrobots with Reinforcement Learning Enhanced by Real-Time ultrasound Imaging Feedback"
Developed a computer vision pipeline for a real-time ultrasound imaging feedback system to control microrobots within the body. Created vein prototypes to demonstrate the concept and control the microrobots.

09.2022–02.2024 **ARC Robotics Hiwi – ETHZ – Zürich, Switzerland**
Assisted the team in improving an automatic waste sorting system by implementing a new computer vision pipeline for waste tracking, developing a simulation of the entire system in Gazebo, and enhancing the presorting stage.
Publication: Esquivel Estay, et al. (2024-04-08). *Dataset and Analysis of River Waste Pollution in the Limmat River, CH, during a One-day City Festival (Zurich Street Parade, 2023)* [White paper].

09.2020–06.2021 **Xplore Robotics Engineer – EPFL – Lausanne, Switzerland**
Responsible for designing and manufacturing the forearm and the gripper of a Mars rover as part of a European inter-university competition.
Awards: - #3 at European Rover Challenge (ERC) for our first year of creation
- Special prize for handling task
- Special prize for science task

Technical Skills

Python	Excellent (IsaacLab, mujoco, pytorch, YOLO, OpenCV)
C/C++	Good (GTKmm)
Matlab	Very good (yalmip, Simulink, OpenCV)
ROS Noetic	Very good
Docker	Good
Javascript	Good
HTML/CSS	Good
Assembler	Good

Catia	Very good
Slodiworks	Very good
Blender	Good
Abaqus	Good

Languages

French	Native language
English	Level C1
German	Level B2

Interests

Downhill/Cross-country skiing
Scuba diving (PADI Open Water)
Climbing
Rap/Rnb music

Reference

Contact information upon request.